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by dense trichomes; Rolling of leaves also help to reduce water uptake by minimizing the transpiring surface; Water storage in succulent tissue of *Agave americana* and other CAM plants have been found to use water conservatively.



Resurrection plants, those plants having ability to survive near total drying which causes them to appear dead. They recover when water is available. Example: *Selaginella lepidophylla*

During drought stress an essential protection mechanism that stabilizes the cell structure is induced gene expression of **stress protein (dehydrin and osmotin)**. These proteins protect the macromolecules in the cytoplasm and in the nucleus, the cytoskeleton (biomembranes) against denaturation. High desiccation tolerance implies that the protoplasm rehydrates when water becomes available. Plants growing in deserts and arid regions are usually drought resistant.

b. Salt Stress

Presence of high salt concentration in the soil restricts the growth and development of plants. Most commonly the plants which are present near the seashore and estuaries are subjected to salt stress. Na^+, Cl^-, K^+, Ca^{++} and Mg^{++} ions usually contribute to soil salinity. Plants growing in such areas face two problems:

1. Absorption of water from the soil with negative water potential
2. Interaction with high concentration of toxic sodium carbonate and chloride ions.

On the basis of salt tolerance, they are grouped into two categories:

1. Halophytes
2. Non-halophytes or glycophytes

Halophytes are native to saline soils. The halophytes which can resist a range of salt concentration are called as **euryhaline** and those with narrow range of resistance are called **stenohaline**. Non-halophytes cannot resist salts as the halophytes. *Helianthus annus* tolerates high Mn^{2+} ions. Those which are present in salt regions face two problems:

- One is high concentration of salts in soil water leads to decrease in water potential so they grow in opposite direction. Example: *Salicornia.*
- Injuries in salt affected plants caused by both osmotic effects and specification effects. Accumulation of chloride ions reduces water absorption and transpiration.

Salt stress due to deficiency of mineral elements (K, P, S, Fe, Mo, Zn, Mg, Mn) causes physiological disorders which lead to reduced growth and yields.

1. Salt accumulators absorb and store salts so that the osmotic potential of their cells continues to remain negative throughout the growing region.
2. In some salt hardy plants, the excess salt is excluded on the surface of leaves. Some plants have salt glands which secrete salt (mostly NaCl). The exuded salt absorbs water hygroscopically from the atmosphere.
3. Some plants lose their excess salt by leaching into the soil or by dropping their salt filled leaves.